

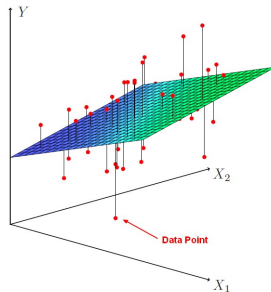
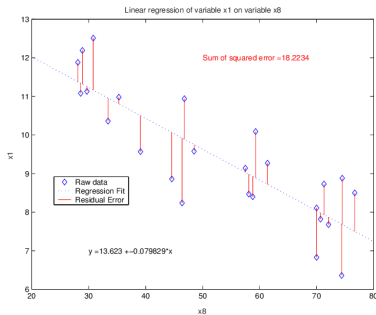


## CSCE 633-700: Machine Learning

Linear regression

# Linear Regression: Evaluation

Minimizing the difference between predicted and actual labels (i.e., prediction error)



1-dimensional input (left), 2-dimensional input (right)

## Linear Regression: Evaluation

- A reasonable thing would be to **minimize the prediction error** (also called residual sum of squares)

$$RSS(\mathbf{w}) = \sum_{n=1}^N (y_n - f(\mathbf{x}_n))^2 = \sum_{n=1}^N \left[ y_n - \left( w_0 + \sum_{d=1}^D w_d x_{nd} \right) \right]^2$$

$x_{nd}$ :  $d^{th}$  feature on the  $n^{th}$  training sample,  $N$  samples,  $D$  features

- An equivalent expression is:  $RSS(\mathbf{w}) = (\mathbf{y} - \mathbf{X}\mathbf{w})^T (\mathbf{y} - \mathbf{X}\mathbf{w})$

$$\mathbf{X} = \underbrace{\begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1D} \\ 1 & x_{21} & x_{22} & \dots & x_{2D} \\ & & \vdots & & \\ 1 & x_{N1} & x_{N2} & \dots & x_{ND} \end{bmatrix}}_{\in \mathbb{R}^{N \times (D+1)}} = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_N^T \end{bmatrix}$$

$$\mathbf{y} = [y_1, \dots, y_N]^T, \mathbf{x}_n = [1, x_{n1}, \dots, x_{nD}]^T, \mathbf{w} = [w_0, w_1, \dots, w_D]^T$$